SAJJADUR RAHMAN

MSc Researcher · Soil Science & Environmental Chemistry

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EDUCATION

Master of Science — Soil Science · University of Dhaka Apr 2025 – Present

CGPA: In progress · National Science and Technology Fellow (Ministry of Science and Technology)

Thesis: Optimizing Nitrogen Efficiency Using Bentonite–Biochar–Coated Urea (BBCU) in Rice and Eggplant Systems

- Core coursework: Advanced Soil Physics · Advanced Soil Chemistry · Soil Carbon Dynamics & Organic Matter Management · Coastal & Wetland Soil Management · Analytical Techniques · Experimental Design · Environmental Biotechnology

Bachelor of Science — Soil, Water, and Environment · University of Dhaka Jan 2020 – Jan 2025

CGPA: 3.60 / 4.00 · Government Scholarship for Academic Excellence

- Relevant coursework: Soil Physics · Pedology · Soil Chemistry · Biostatistics · Ecology · GIS & Remote Sensing · Hydrology · Plant Nutrition · Climate Change Mitigation & Adaptation · Agronomy · Soil Microbiology
- Seminar paper: Internet of Things (IoT) in the Development of Agriculture 5.0: Prospects and Challenges
- Laboratory training: soil morphology, environmental chemistry, GIS/remote sensing, biostatistics (R), hydrology, petrology & mineralogy

RESEARCH EXPERIENCE

MSc Thesis Researcher · University of Dhaka May 2025 – Present

Optimizing Nitrogen Efficiency Using Bentonite–Biochar–Coated Urea in Rice and Eggplant Systems

- Designing CRD-based pot and field experiments (8 treatments × 3 replications) evaluating biochar–bentonite-coated urea as a climate-smart slow-release fertilizer for South Asian rice–vegetable systems.
- Quantifying four NUE indices — Agronomic Efficiency (AE), Recovery Efficiency (RE), Physiological Efficiency (PE), and Nitrogen Harvest Index (NHI) — against an unfertilized control.
- Post-harvest soil analysis: residual NH_4^+/NO_3^- (KCl extraction), available nutrients, Walkley-Black OC, pH, and EC.
- Modelling N release from BBCU with zero-order, first-order, Higuchi and Korsmeyer–Peppas kinetics.
- Developing field-scalable, climate-smart fertilizer management strategies for Bangladesh's rice–vegetable agroecological conditions.
- Awarded National Science and Technology Fellowship (Ministry of Science and Technology, Bangladesh) for this research.

Research Assistant · Dhaka University Nanotechnology Centre (DUNTC) Apr 2025 – Apr 2026

Role of Nano-Fertilizers on Growth, Yield, and Nutrient Quality of Wheat (BARI Gom 33)

- Conducted a CRD pot experiment (8 treatments × 3 replications, 24 pots) comparing nano-urea versus conventional urea application methods on wheat growth and soil health.
- After hail damage, redirected focus to plant tissue nutrient quality (Kjeldahl N, protein estimation at N% × 5.7, P/K/S/Ca/B via AAS and ICP-OES) and NUE indices.
- Post-harvest soil dynamics: residual mineral N, available nutrients, pH, EC, and Walkley-Black organic carbon to assess fertilizer carry-over effects.
- Developed SPADE (Statistical Platform for Agronomic Data Evaluation) — an open-source Python/Streamlit application for integrated NUE computation, factorial ANOVA with Tukey HSD, outlier detection, and publication-ready figure export; manuscript in preparation.
- Managed a master sample log covering 24-pot tissue and soil datasets; prepared project completion report for DUNTC.

Research Trainee — Atmospheric Pollution & Metal Analysis · BCSIR Nov 2025 – Present

- Conducting size-resolved particulate matter sampling ($PM_{0.1}$, PM_1, $PM_{2.5}$) at three contrasting sites in the Dhaka region (Dholaikhal, Savar, [third site]) using Nanosampler II (Model 3182, Kanomax, Japan).
- Developed filter digestion SOPs (EN 14902:2005 microwave acid digestion, HNO_3 + H_2O_2) for glass-fibre/PTFE filters, with QA/QC via NIST SRM 1648a, method blanks and spike recovery.
- Characterising heavy metals by ICP-MS and AAS (Varian AA 240 FS) to assess atmospheric contamination and ecological risk indices; manuscript in preparation.

Independent Researcher — Environmental Governance · University of Dhaka 2026 – Present

The Political Economy of Environmental Governance Failure in Bangladesh

- Built a district-level Environmental Governance Failure Index (EGFI) for Bangladesh and tested whether governance burden shifted after the July 2024 political transition; district rankings were near-identical before and after (Spearman $\rho = 0.998$; ICC = 0.998).
- Modelled spatial determinants with OLS, a spatial error model and geographically weighted regression (GWR $R^2 = 0.56$ vs. OLS $R^2 = 0.10$); released a six-script reproducibility package.
- Presented at ICJR-II (July 2026); single-author manuscript submitted to *Global Environmental Change* (October 2026).

Legacy Lead Contamination at a Decommissioned Battery Recycling Site 2026 – Present

Fulbaria, Narsingdi · Department of Soil, Water and Environment, University of Dhaka | Supervisor: [name]

- Designed and carried out a field sampling campaign of soils, drainage sediment and pond water across a source-to-farmland gradient.
- Measured 13 trace elements by ICP-MS, along with soil pH, EC, organic carbon and texture.
- Assessed contamination and risk using geoaccumulation, enrichment, pollution load and ecological risk indices, a distance-decay model, and a USEPA human health risk assessment.
- Ongoing: sequential extraction (BCR) and remediation experiments with biochar and phosphate amendments.

Research Assistant · Department of Soil, Water & Environment, University of Dhaka 2025 – Present

Phytoremediation Efficiency of Indoor Plants in Optimizing Indoor Air Quality

- Assisting with study design and methodology development for evaluating the phytoremediation potential of selected indoor plant species under controlled indoor conditions.
- Contributed to plant procurement and transportation to experimental sites; assisted with air quality monitoring, data collection, and will contribute to analysis and reporting through project completion.
- Quantified leaf ascorbic acid in six indoor plant species (colorimetric, 520 nm; 17-point calibration, $R^2 = 0.9967$); manuscript in preparation.

Principal Investigator (Self-Funded) · Independent Research Team, Dhaka Jan 2025

Integrated Assessment of Heavy Metal Contamination in Industrial Areas of Dhaka

- Led a four-member team collecting and analysing 40+ soil, dust, and leaf samples from industrial zones surrounding Dhaka.
- Assessed spatial distribution, contamination indices (geo-accumulation index, enrichment factor), and ecological risk factors of heavy metals affecting surrounding agro-ecosystems.

Graduate Seminar Researcher · University of Dhaka Jun 2024 – Sep 2024

- Reviewed 60+ peer-reviewed publications on IoT applications in Agriculture 5.0; presented evidence-based recommendations to faculty and peers on smart farming technologies for Bangladesh.

Undergraduate Research Group Member · University of Dhaka Jan 2023 – Nov 2025

- Collaborated with a 3–5-member team on sustainable agriculture and climate resilience projects.
- Conducted socio-economic surveys on organic fertilizer adoption in Dhaka's urban/peri-urban farming systems; manuscript submitted to SAARC Journal of Agriculture (October 2026).
- Presented findings at four international conferences (2024–2025).

PUBLICATIONS & SOFTWARE

Submitted Manuscripts

1. Rahman, S. *The Political Economy of Environmental Governance Failure: Spatial Determinants, Temporal Persistence, and the Limits of Political Change in Bangladesh*. Submitted — **Global Environmental Change** (Elsevier), October 2026.
2. Rahman, S., Hossain Rubab, M. M., Payal, M. P., Maksud, F. H., & Islam, R. *Organic Fertilizer Adoption Completeness Among Urban Growers in Dhaka: Site Type as a Determinant of Exclusive Versus Mixed Organic-Synthetic Practice*. Submitted — **SAARC Journal of Agriculture** (SAARC Agriculture Centre), October 2026.

Manuscripts in Preparation

1. Rahman, S., [co-authors] *Nitrogen release kinetics and use efficiency of bentonite–biochar-coated urea (BBCU) as a slow-release fertilizer*. [working title] In preparation.
2. Rahman, S., [co-authors] *Size-resolved heavy-metal composition of particulate matter ($PM_{0.1}$, PM_1, $PM_{2.5}$) at three sites in the Dhaka region*. [working title] In preparation.
3. Rahman, S., [co-authors] *Legacy lead contamination at a decommissioned battery recycling site, Fulbaria, Narsingdi, Bangladesh*. [working title] In preparation.
4. [Authors — confirm order incl. Rahman, S.] *Phytoremediation efficiency of indoor plants in improving indoor air quality*. [working title] In preparation.
5. Rahman, S. *SPADE: An Integrated Open-Source Framework for Standardised Nitrogen Use Efficiency Analysis in Small-Plot Agricultural Experiments*. Target: Computers and Electronics in Agriculture (Elsevier) / SoftwareX.

Software

- **SPADE v1.0** — Statistical Platform for Agronomic Data Evaluation. Open-source Python/Streamlit application for NUE computation (five indices), factorial ANOVA with Tukey HSD and compact letter display, outlier detection (Dixon's Q, Grubbs), and publication-ready PNG/JPG figure export. GitHub: github.com/sajjadurrahmanzadid-hue.

Conference Presentations & Proceedings

1. Rahman, S. (2026, July). *The political economy of environmental governance failure: Spatial determinants, temporal persistence, and the limits of political change in Bangladesh.* 2nd International Conference on the July Revolution (ICJR-II), University of Dhaka, Bangladesh.
2. Rahman, S. (2025, September). *Nano-fertilizers in wheat cultivation: A comprehensive review and strategic framework for sustainable productivity and biofortification.* https://doi.org/10.13140/RG.2.2.29613.35043
3. Rahman, S., Fairuz, A., & Tahiyat, T. (2024, December). *Economic viability and sustainability of hydroponics in flood-prone Bangladesh.* Asian Institute of Technology International Conference, Bangkok, Thailand.
4. Rahman, S. (2024, December). *Internet of Things (IoT) on the development of Agriculture 5.0: Prospects and challenges.* In G. N. Günay (Ed.), Proceedings of the 3rd International Ege Congress on Scientific Research (pp. 170–178). IKSAD Publishing House. https://doi.org/10.13140/RG.2.2.17269.69600
5. Rahman, S., Hossain Rubab, M. M., Payal, M. P., Maksud, F. H., & Islam, R. (2024, June). *Climate change and resilient farming approaches in Bangladesh: A review of potential impacts and adaptation strategies.* In Proceedings of the International Congress on Food, Agriculture and Environmental Research in the Global World. https://doi.org/10.13140/RG.2.2.33718.72005
6. Rahman, S., Hossain Rubab, M. M., & Payal, M. P. (2024, May). *Developing a smart, sustainable agriculture initiative in Bangladesh: A data-driven approach to ensuring food security and minimizing environmental impact.* In Proceedings of the HODJA AKHMET YASSAWI 8th International Congress on Scientific Research. https://doi.org/10.13140/RG.2.2.35396.44165
7. Rahman, S., Hossain Rubab, M. M., Payal, M. P., Maksud, F. H., & Islam, R. (2024, March). *Navigating sustainability in urban landscapes: Assessing organic fertilizer adoption and challenges among farmers in the Dhaka region, Bangladesh.* In Proceedings of the International Congress on Sustainable Agriculture. https://doi.org/10.13140/RG.2.2.21075.55849
8. Rahman, S., Hossain Rubab, M. M., & Payal, M. P. (2023, May). *Improving crop productivity in Bangladesh through advanced yield prediction techniques.* In Proceedings of the IEEE SIGHT Student Branch, University of Dhaka "Ideas for Life" Conference. https://doi.org/10.13140/RG.2.2.16907.36644
9. Rahman, S., Hossain Rubab, M. M., & Payal, M. P. (2022, November). *An assessment and analysis of Bangladesh's agricultural approach based on technology.* In Proceedings of the 2nd International Competition for Young Researchers 2022. https://doi.org/10.13140/RG.2.2.13587.96809

TECHNICAL SKILLS

Instrumentation: Nanosampler II (Kanomax Model 3182) · ICP-MS · AAS (Varian AA 240 FS) · ICP-OES (multi-element tissue analysis) · microwave digestion · flame photometry

Analytical Methods: Kjeldahl N digestion & distillation · Protein estimation (N% × 5.7) · Walkley-Black OC · Indophenol blue colorimetry (NH_4^+) · KCl soil extraction · EC/pH measurement · Contamination indices (geo-accumulation, enrichment factor, pollution load, ecological risk) · USEPA human health risk assessment · BCR sequential extraction · PM filter acid digestion (EN 14902; NIST SRM 1648a QA/QC) · Bouyoucos particle size · NH_4OAc CEC · colorimetric ascorbic acid

Experimental Design: CRD · RCBD · pot and field experiments · leaching columns · jar-based volatilisation · NUE index calculation (AE, RE, PE, NHI, PFP) · N release kinetics modelling (zero-order, first-order, Korsmeyer-Peppas, Higuchi)

Computing & Data: Python (Streamlit, pandas, scipy, matplotlib, statsmodels — developer of SPADE) · R (biostatistics, ANOVA, regression, Firth penalised logistic regression) · Excel (advanced) · SPSS · Minitab · Git/GitHub

Spatial Analysis: QGIS · remote sensing · ArcGIS (basic) · spatial econometrics (Moran's I, spatial error models, GWR)

Laboratory: Soil & water chemistry · soil morphological analysis · microbiological analysis · hydrology · heavy metal speciation

Scientific Communication: Zotero · scientific writing · Wiley Researcher Academy certified · Stanford Writing in the Sciences certified · manuscript preparation for international journals

Software & Tools: Microsoft Office · Google Suite · LaTeX (basic) · Streamlit (dashboard development)

AWARDS & FELLOWSHIPS

2026 National Science and Technology Fellowship — Ministry of Science and Technology, Bangladesh

2026 Government Scholarship for BSc Academic Excellence — Government of Bangladesh

2025 Research Fellowship — Dhaka University Nanotechnology Centre (DUNTC)

2025 Transforming Dhaka into an Eco-Friendly Megacity — Preliminary Round Winner, Ministry of Youth & Sports

2023 Project GreenZen: Seaweed-Based Biodegradable Packaging — Top 10% of 50 competing teams

2022 Gold Award — 2nd International Competition for Young Researchers (top 10%)

2020 Jahangir Alam Education Foundation Scholarship

2017 & 2019 Talent Pool Scholarships — SSC 2017 · HSC 2019

LEADERSHIP & ACADEMIC SERVICE

President · Soil Club – University of Dhaka Chapter Nov 2025 – Present

- Conceptualised and coordinated the 'Healthy Soils for Healthy Cities' World Soil Day 2025 campaign — 50,000+ outreach; delivered a technical presentation on organic fertilizer adaptation for urban soil sustainability to an international audience.

Class Representative · MS Session 2023-24, University of Dhaka Mar 2025 – Present

- Representing 120+ students; coordinating between students, faculty, and administration; assisting professors with scheduling and departmental communications.

Magazine Editor · Byapon – Youth Science Magazine Jan 2022 – Dec 2024

- Edited and authored science content; led editorial and design strategy reaching 50,000+ youth readers in Bangladesh.

Research Intern · UniV Dec 2021 – Sep 2024

- Organised 30+ workshops on research methods, R, bioinformatics, IELTS, and Python for 500+ students; collaborated with 20+ researchers on academic and environmental seminars.

Student Mentor & Counsellor · University of Dhaka Jan 2021 – Present

- Mentored 100+ students in academic guidance, research skills, and idea development.

Campus Research Co-Ambassador · Be Researcher World Forum Mar 2021 – Mar 2022

- Coordinated 15+ webinars on environmental/agricultural research for 100+ students; led the graphics and design team.

Executive Secretary · Dhaka University Model OIC Club Feb 2020 – Feb 2022

- Managed research/public relations teams and communications for 50+ members; oversaw academic and environmental policy-related student projects.

PROFESSIONAL DEVELOPMENT

Dec 2025 Key Components of a Research Article — Wiley Researcher Academy

Aug 2025 Writing in the Sciences — Stanford University (Coursera)

Sep 2024 Green Skills and Social Accountability — Global Youth Leadership Center

Jan 2024 Introduction to R — UniV

Jan 2021 Research Methodology — Be Researcher World Forum

LANGUAGES

English — Proficient. IELTS 7.5: Listening 8.0 · Reading 8.0 · Speaking 7.5 · Writing 6.5

Bangla — Native

REFERENCES

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